

Advanced (Carolina Reaper)

Q1. Factor $x^2 - 4$

- (A) $(x + 2)(x - 2)$
- (B) $x(x - 4)$
- (C) $x(x + 4)$
- (D) $x^2 - 4x$
- (E) Cannot be factored

Q2. Factor $49 - y^2$

- (A) $(7 + y)(7 - y)$
- (B) $(7 - y)(7 + y)$
- (C) $(7 + y)^2$
- (D) $49y^2$
- (E) Cannot be factored

Q3. Simplify $(a^2 - b^2) + (b^2 - c^2)$

- (A) $a^2 - c^2$
- (B) $b^2 - a^2$
- (C) $c^2 - a^2$
- (D) $a^2 + b^2 - c^2$
- (E) $a^2 - b^2 + c^2$

Q4. Factor $9x^2 - 16y^2$

- (A) $(3x + 4y)(3x - 4y)$
- (B) $(3x - 4y)(3x + 4y)$
- (C) $(3x)^2 - (4y)^2$
- (D) $9x^2 - 16$
- (E) Cannot be factored

Q5. Factor $x^2 - 9y^2$

- (A) $(x - 3y)(x + 3y)$
- (B) $(x + 3y)(x - 3y)$
- (C) $x^2 - 9$
- (D) $x^2 - y^2$
- (E) Cannot be factored

Q6. Solve $(x^2 - 4)(x + 2) = 0$

- (A) $x = -2, x = 2$
- (B) $x = -2, x = -1$
- (C) $x = 2, x = 1$
- (D) $x = -2$
- (E) $x = 0$

Q7. Factor $x^4 - y^4$

- (A) $(x^2 + y^2)(x^2 - y^2)$
- (B) $(x^2 - y^2)(x^2 + y^2)$
- (C) $x^4 - y^4$
- (D) $x^2 - y^2$
- (E) Cannot be factored

Q8. Factor $16x^2 - 9y^2$

- (A) $(4x - 3y)(4x + 3y)$
- (B) $(4x + 3y)(4x - 3y)$
- (C) $16x^2 - 9y$
- (D) $16x - 9y$
- (E) Cannot be factored

Open Ended Questions (FRQ)

Q9. Prove that $(a + b)(a - b) = a^2 - b^2$. Show all steps and explain your reasoning.

Q10. Factor $x^2 - 8x + 16 - y^2$. Simplify your answer and explain your steps.

Chef's Tips

- Remind students to apply the difference of squares formula
- Encourage students to check their work and simplify their answers
- Provide examples of each type of problem before the test